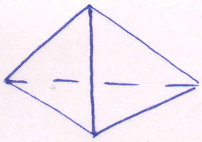
№12.



$H_0: \begin{cases} 1/4 \\ 1/4 \\ 1/6 \\ 1/3 \end{cases}$ $H_1: \begin{cases} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{cases}$

	(1,1)	(1,2)	(1,3)	(1,4)	(2,2)	(2,3)	(2,4)	(3,3)	(3,4)	(4,4)
$H_0: P_0$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{16}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{36}$	$\frac{1}{9}$	$\frac{1}{9}$
$H_1: P_1$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$
$\ell = L_1/L_0$	1	1	$\frac{3}{2}$	$\frac{3}{4}$	1	$\frac{3}{2}$	$\frac{3}{4}$	$\frac{9}{4}$	$\frac{9}{8}$	$\frac{9}{16}$

$P(\ell \geq c \mid H_0) \leq \alpha = 0{,}2 = 2/10$

для $c = \frac{9}{4}$, $\frac{1}{36} < 0{,}2$

для $c = \frac{3}{2}$ $\frac{1}{36} + \frac{1}{12} + \frac{1}{12} = 0{,}194\ldots < 0{,}2$

для $c = \frac{9}{8}$ $\frac{1}{36} + \frac{1}{12} + \frac{1}{12} + \frac{1}{36} = 0{,}222\ldots > 0{,}2$

$\Rightarrow c = 3/2$

G: появление (1,3), (2,3), (3,3)

$W = P(\vec{x} \in G \mid H_1) = P(\ell \geq \frac{3}{2} \mid H_1) = \frac{1}{16} + \frac{1}{8} + \frac{1}{8} = \frac{5}{16}$